

NOVATECH

HR Attrition Analytics Dashboard

DATA DICTIONARY & DATA GOVERNANCE REFERENCE

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This document defines the analytical dataset used in the NovaTech HR Attrition Analytics Dashboard, including field definitions, permitted values, derived logic, data-quality controls, and governance notes.

Document Control

Property	Value
Project	HR Attrition Analytics Dashboard
Company	NovaTech (fictional)
Department	Human Resources
Dataset	Synthetic employee attrition dataset
Tool	Microsoft Excel
Raw records	1,488
Analysis-ready records	1,468
Analytical fields	37
Primary outcome	Attrition
Owner	Amirreza Akbari

Revision History

Version	Date	Author	Change
0.1	July 2026	Amirreza Akbari	Initial compilation from workbook dictionary
0.9	July 2026	Amirreza Akbari	Definitions, business use, and validation rules added
1.0	July 2026	Amirreza Akbari	Final portfolio release

How to Use This Document

Use this reference when reviewing the workbook, validating calculations, reproducing PivotTables, interpreting dashboard visuals, or extending the analysis. The in-workbook Data_Dictionary sheet remains a concise quick reference; this PDF is the comprehensive governance version.

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1. Dataset Overview

The analytical dataset supports descriptive HR analytics focused on employee turnover. It combines employee demographics, organizational placement, compensation, work conditions, satisfaction indicators, career history, and tenure variables. The dataset is designed for executive reporting rather than predictive modeling.

Layer	Count	Purpose
Raw data	1,488 records / 35 columns	Preserves source data and data-quality issues
Cleaned data	1,468 records / 37 columns	Validated analysis-ready dataset with derived fields
Data dictionary	37 documented fields	Defines meaning, types, values, and business use

Primary Analytical Outcome

Attrition indicates whether an employee left the organization. AttritionFlag converts the outcome to 1/0 so PivotTables and KPI calculations can aggregate employee exits reliably.

2. Data Architecture and Lineage

The workbook follows a controlled flow from raw source data to cleaned analysis, PivotTables, dashboard reporting, written insights, and recommendations.

- 1. Raw_Data - preserved source layer**
- 2. Cleaned_Data - standardized and validated layer**
- Pivot_Tables - aggregated analysis layer
- Dashboard - executive visualization layer
- Insights - evidence-based interpretation
- Recommendations - action-oriented HR response

Data Lineage Principle

Raw data should not be overwritten. All cleaning decisions and derived fields belong in the cleaned layer. Dashboard metrics must reconcile to the cleaned dataset and PivotTables.

3. Complete Field Dictionary

The following table documents all 37 analytical fields. Derived fields are explicitly identified to separate source attributes from analyst-created segmentation and measures.

Field	Type	Definition	Category	Allowed values / rule	Business use
Age	Numeric	Employee age in years (18-60)	Demographics	18-60	Workforce profile and age-risk analysis
AgeGroup	Categorical	Derived band: Under 25 / 25-34 / 35-44 / 45-54 / 55+	Derived segmentation	Under 25; 25-34; 35-44; 45-54; 55+	Compare attrition across career stages
Attrition	Categorical	Whether the employee has left the company (Yes/No)	Outcome	Yes; No	Primary target variable for turnover analysis
AttritionFlag	Numeric	1 if Attrition = Yes, 0 if No (for summing in pivots)	Derived measure	0; 1	Supports attrition counts and rate calculations
BusinessTravel	Categorical	Employee travel frequency	Work conditions	Non-Travel; Travel_Rarely; Travel_Frequently	Assess whether travel intensity relates to attrition
DailyRate	Numeric	Daily pay rate	Compensation	Positive numeric value	Compensation distribution and pay-pattern analysis
Department	Categorical	Organizational department	Organization	Sales; Research & Development; Human Resources	Compare turnover across business functions
DistanceFromHome	Numeric	Commute distance from home to workplace	Work conditions	Non-negative numeric value	Evaluate commute burden as a retention factor
Education	Numeric	Education level code, 1 (lowest) to 5 (highest)	Demographics	1-5	Profile workforce education and compare attrition
EducationField	Categorical	Employee field of study	Demographics	Life Sciences; Medical; Marketing; Technical Degree; Human Resources; Other	Identify education-background patterns
EmployeeNumber	Numeric / ID	Unique employee identifier	Identifier	Unique positive integer	Record integrity and employee counting
EnvironmentSatisfaction	Numeric	Satisfaction with the work environment, 1-4	Employee experience	1-4	Measure workplace environment risk
Gender	Categorical	Employee gender; Unknown indicates missing source value	Demographics	Male; Female; Unknown	Demographic comparison and data-quality transparency
HourlyRate	Numeric	Hourly pay rate	Compensation	Positive numeric value	Analyze pay distribution
JobInvolvement	Numeric	Job involvement score, 1-4	Employee experience	1-4	Assess engagement and commitment
JobLevel	Numeric	Job seniority level, 1-5	Organization	1-5	Compare attrition by seniority
JobRole	Categorical	Specific employee job title	Organization	Role values defined in source data	Identify high-risk roles
JobSatisfaction	Numeric	Job satisfaction score, 1-4	Employee experience	1-4	Analyze satisfaction as a retention factor
SatisfactionGroup	Categorical	Derived label from JobSatisfaction	Derived segmentation	Low; Medium; High; Very High	Create executive-friendly satisfaction categories
MaritalStatus	Categorical	Employee marital status	Demographics	Single; Married; Divorced	Demographic attrition comparison
MonthlyIncome	Numeric	Monthly salary	Compensation	Positive numeric value	Assess compensation and attrition patterns
IncomeGroup	Categorical	Derived monthly-income band	Derived segmentation	Low Income; Middle Income; High Income	Compare turnover across salary bands
MonthlyRate	Numeric	Separate internal monthly pay metric	Compensation	Positive numeric value	Supplementary compensation analysis
NumCompaniesWorked	Numeric	Number of companies worked at before NovaTech	Career history	Non-negative integer	Assess job-mobility patterns
OverTime	Categorical	Whether the employee regularly works overtime	Work conditions	Yes; No	Measure workload-related attrition risk
PercentSalaryHike	Numeric	Most recent salary increase percentage	Compensation	Percentage value	Assess relationship between salary progression and retention
PerformanceRating	Numeric	Performance rating code	Performance	3-4	Compare retention across performance levels
RelationshipSatisfaction	Numeric	Satisfaction with workplace relationships, 1-4	Employee experience	1-4	Assess interpersonal workplace experience
StockOptionLevel	Numeric	Stock option level, 0-3	Compensation	0-3	Evaluate long-term incentive relationship with retention
TotalWorkingYears	Numeric	Total years of professional experience	Career history	Non-negative integer	Compare attrition by career experience
TrainingTimesLastYear	Numeric	Number of training events attended last	Development	Non-negative integer	Assess learning investment and

Field	Type	Definition	Category	Allowed values / rule	Business use
ar		year			retention
WorkLifeBalance	Numeric	Work-life balance score, 1 (bad) to 4 (best)	Employee experience	1-4	Evaluate well-being and retention
YearsAtCompany	Numeric	Employee tenure at NovaTech in years	Tenure	Non-negative integer	Measure early- and long-tenure attrition
TenureGroup	Categorical	Derived tenure band	Derived segmentation	0-1 Years; 2-4 Years; 5-9 Years; 10+ Years	Create executive-friendly tenure comparisons
YearsInCurrentRole	Numeric	Years in current role	Tenure	Non-negative integer	Assess role stagnation and movement
YearsSinceLastPromotion	Numeric	Years since last promotion	Career progression	Non-negative integer	Evaluate promotion timing and attrition
YearsWithCurrManager	Numeric	Years working with current manager	Management relationship	Non-negative integer	Assess manager continuity and retention

4. Coded Value Definitions

Education

Code	Interpretation
1	Below College
2	College
3	Bachelor
4	Master
5	Doctor

Satisfaction / Involvement Scores

Code	Interpretation
1	Low
2	Medium
3	High
4	Very High

Work-Life Balance

Code	Interpretation
1	Bad
2	Good
3	Better
4	Best

Performance Rating

Code	Interpretation
3	Meets expectations
4	Exceeds expectations

Stock Option Level

Code	Interpretation
0	No stock option
1	Level 1
2	Level 2
3	Level 3

Job Level

Code	Interpretation
1	Entry / junior
2	Developing professional
3	Experienced / specialist
4	Senior / management
5	Executive / highest level

Categorical Values

Field	Permitted values
Attrition	Yes; No
BusinessTravel	Non-Travel; Travel_Rarely; Travel_Frequently
Department	Sales; Research & Development; Human Resources
Gender	Male; Female; Unknown
MaritalStatus	Single; Married; Divorced
OverTime	Yes; No
SatisfactionGroup	Low; Medium; High; Very High
IncomeGroup	Low Income; Middle Income; High Income
TenureGroup	0-1 Years; 2-4 Years; 5-9 Years; 10+ Years
AgeGroup	Under 25; 25-34; 35-44; 45-54; 55+

5. Derived Field Logic

Derived field	Business rule	Analytical purpose
AttritionFlag	IF Attrition = "Yes" THEN 1 ELSE 0	Enables summation and calculation of attrition count and rate.
AgeGroup	Age <25; 25-34; 35-44; 45-54; 55+	Creates life-stage and career-stage groups.
IncomeGroup	MonthlyIncome <3,000 = Low; 3,000-6,999 = Middle; >=7,000 = High	Supports salary-band comparisons.
TenureGroup	YearsAtCompany 0-1; 2-4; 5-9; 10+	Identifies early-tenure and long-tenure risk.
SatisfactionGroup	1=Low; 2=Medium; 3=High; 4=Very High	Translates coded score into executive-friendly labels.

Metric Logic

Metric	Definition
Total Employees	Count of unique EmployeeNumber
Attrition Count	Sum of AttritionFlag
Active Employees	Total Employees - Attrition Count
Attrition Rate	Attrition Count / Total Employees
Average Monthly Income	Average of MonthlyIncome
Average Age	Average of Age
Average Tenure	Average of YearsAtCompany

6. Data Quality and Validation Rules

Validation	Rule	Status
Record uniqueness	EmployeeNumber must be unique in the cleaned dataset.	Passed
Missing critical fields	EmployeeNumber, Attrition, Department, JobRole, Age, MonthlyIncome must be populated.	Passed
Attrition domain	Only Yes or No is accepted.	Passed
Numeric ranges	Age, satisfaction scores, education, job level, and performance must remain within defined ranges.	Passed
Derived field consistency	Derived groups must match their source values and documented thresholds.	Passed
Raw-to-clean reconciliation	Removed or corrected records must be explainable through the cleaning process.	Passed
KPI reconciliation	Dashboard KPIs must match PivotTable totals and formulas.	Passed

Data-quality note: Gender may contain the category Unknown to preserve transparency where the source record was missing. This value should not be silently imputed without a defensible business rule.

7. Usage and Governance Notes

Raw data preservation: Do not edit or delete the Raw_Data sheet. It serves as the traceable source layer.

Cleaned data ownership: All corrections, standardization, and helper fields belong in Cleaned_Data.

Definition control: Any threshold change for AgeGroup, IncomeGroup, TenureGroup, or SatisfactionGroup must be updated in both the workbook and this document.

Interpretation limits: The analysis identifies associations, not causation. Attrition patterns should be validated with HR context before policy changes.

Privacy: The dataset is synthetic and intended for portfolio demonstration. Real HR data would require access control, minimization, and privacy review.

Refresh procedure: When replacing the dataset, rerun uniqueness, missing-value, domain, and KPI reconciliation checks before refreshing the dashboard.

8. Glossary

Term	Definition
Attrition	An employee leaving the organization.
Categorical variable	A field that represents a label or group rather than a continuous number.
Derived field	A new analytical field created from one or more source variables.
KPI	Key Performance Indicator used to monitor business performance.
PivotTable	Excel summary structure used to aggregate and compare data.
Data lineage	The documented path from source data through transformation to reporting.
Validation rule	A condition used to confirm that data is complete, accurate, and consistent.

End of document